

GenAI Retail Analytics Assistant

Transforming Raw Data into Real-Time Insights
Tech: Python | Gemini 2.5 Flash | LangGraph | Streamlit

System Architecture

- Streamlit UI (Chat + Dashboard)
- LangGraph Orchestration
 - Resolver Agent
 - Code Agent
 - Validator Agent
- Gemini 2.5 Flash LLM
- Unified Retail Data Schema

Scalability Strategy (100GB+ Data)

- Current Prototype:
 - CSV + Pandas (10GB limit)
- Production Approach:
 - Delta Lake
 - PySpark Distributed Compute
 - RAG + Text-to-SQL
 - Redis Query Caching

Live Demo Evidence(Executive Summary)



Retail Analyst AI

 **Modes**

Select Mode

☐ Conversational Q&A

☒ Executive Summary

 **Tip:** Connects to Gemini 2.5 Flash for sub-second analytics.

Clear Conversation



Interactive Retail Data Assistant

 **Executive Performance Report** 

Generate Live Summary

 **AI-Generated Report**

Executive Summary: Sales Performance, Top Categories, and Risks

Date: October 26, 2023

Prepared For: Executive Leadership Team

Overview: Our sales performance demonstrates robust top-line growth, achieving significant revenue and transaction volumes. However, a deeper dive into profitability and financial health reveals critical data inconsistencies that require immediate attention to accurately assess our true financial standing and mitigate potential risks.

1. Sales Performance Highlights

The company has achieved strong sales figures, indicating healthy market demand and effective sales strategies:

- **Total Sales Revenue:** A substantial \$78,592,678.30 has been generated.
- **Total Quantity Sold:** We moved 116,649 units, reflecting strong product uptake.

Conversational Q&A Engine



Retail Analyst AI

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Clear Conversation

Deploy 



Interactive Retail Data Assistant

 Hello! I have access to your Sales, Inventory, and Financial data. Ask me anything!

 which region has the highest cancellation rate

 Based on the data provided:

The region with the highest cancellation rate is **APO**.

This finding indicates that the Asia-Pacific Operations (APO) region experiences the highest proportion of cancellations. As a Business Analyst, I would recommend further investigation into the specific factors contributing to this high cancellation rate in APO. This could involve analyzing customer feedback, sales processes, product/service issues, competitive landscape, or market conditions unique to the APO region to identify root causes and potential solutions.

 which region has the lowest sale

 Based on the data result provided:

Ex: 'Which region has the highest cancellation rate?' 

Conclusion & Future Scope

- Business Value:
 - Democratizes Data Access
 - Sub-Second Insights
 - Unified Retail Intelligence
- Next Steps:
 - Forecasting Models
 - Real-Time Streaming